

# FLEX NORYL™ RESIN WCD855

REGION ASIA

## DESCRIPTION

Flexible and non-halogenated flame retardant extrusion grade intended for evaluation in applications such as cable jacket of UL 62 SPE and NISPE configurations. Flame retardant performance capable of meeting UL 1581 VW-1 requirement. 90C or 105C temperature rating as defined by UL 62 TPE category. 85 Shore A hardness. Processing typically conducted on standard extrusion equipment. UL 1581 tests conducted on 2.0 mm wire with 0.12 mm x 20 stranded copper conductor.

## TYPICAL PROPERTY VALUES

Revision 20170913

| PROPERTIES                                        | TYPICAL VALUES | UNITS    | TEST METHODS   |
|---------------------------------------------------|----------------|----------|----------------|
| <b>MECHANICAL</b>                                 |                |          |                |
| Tensile Stress, brk, Type I, 50 mm/min            | 16             | MPa      | ASTM D 638     |
| Tensile Strain, brk, Type I, 50 mm/min            | 260            | %        | ASTM D 638     |
| Flexural Modulus, 12.5 mm/min, 100 mm span        | 60             | MPa      | ASTM D 790     |
| Hardness, Shore A, 30S reading                    | 85             | -        | ASTM D 2240    |
| Tensile Stress, break, 50 mm/min                  | 16             | MPa      | ISO 527        |
| Tensile Strain, break, 50 mm/min                  | 250            | %        | ISO 527        |
| Flexural Modulus, 12.5 mm/min                     | 40             | MPa      | ISO 178        |
| <b>IMPACT</b>                                     |                |          |                |
| Brittleness Temperature                           | <-40           | °C       | ASTM D 746     |
| <b>PHYSICAL</b>                                   |                |          |                |
| Specific Gravity                                  | 1.01           | -        | ASTM D 792     |
| Melt Flow Rate, 250°C/10.0 kgf                    | 16             | g/10 min | ASTM D 1238    |
| <b>ELECTRICAL</b>                                 |                |          |                |
| Volume Resistivity                                | 1.9E+16        | Ohm-cm   | ASTM D 257     |
| Relative Permittivity, 1 MHz                      | 2.4            | -        | ASTM D 150     |
| Dissipation Factor, 1 MHz                         | 0.007          | -        | ASTM D 150     |
| Dielectric strength in oil, 2.0mm                 | 25             | kV/mm    | IEC 60243-1    |
| Comparative Tracking Index                        | 600            | V        | IEC 60112      |
| <b>FLAME CHARACTERISTICS</b>                      |                |          |                |
| Smoke Density on 0.5mm plaque, Non-flame, Ds, max | 47             | -        | ASTM E 662     |
| Smoke Density on 0.5mm plaque, Flame, Ds, max     | 110            | -        | ASTM E 662     |
| Glow Wire Flammability Index 850°C, passes at     | 3              | mm       | IEC 60695-2-12 |

| PROPERTIES                                                                          | TYPICAL VALUES | UNITS | TEST METHODS   |
|-------------------------------------------------------------------------------------|----------------|-------|----------------|
| Glow Wire Ignitability Temperature, 3.0 mm                                          | 825            | °C    | IEC 60695-2-13 |
| Oxygen Index (LOI)                                                                  | 26             | %     | ISO 4589       |
| <b>WIRE AND CABLE - UL 1581 TESTED ON 2.0MM WIRE WITH 0.12MMX20 STRANDED COPPER</b> |                |       |                |
| Tensile strength @ break                                                            | 23             | MPa   | UL 1581        |
| Tensile elongation @ break                                                          | 308            | %     | UL 1581        |
| Tensile strength @ break after 7days @136°C                                         | 22             | MPa   | UL 1581        |
| Tensile elongation @ break after 7days @136°C                                       | 251            | %     | UL 1581        |
| Heat Deformation at 121°C/250g                                                      | 5              | %     | UL 1581        |
| VW-1                                                                                | Pass           | -     | UL 1581        |
| <b>WIRE COATING EXTRUSION</b>                                                       |                |       |                |
| Drying Temperature                                                                  | 75 – 85        | °C    |                |
| Drying Time                                                                         | 5 – 7          | hrs   |                |
| Drying Time (Cumulative)                                                            | 12             | hrs   |                |
| Maximum Moisture Content                                                            | 0.02           | %     |                |
| Extruder Length/Diameter Ratio (L/D)                                                | 22:1 to 26:1   | -     |                |
| Screw Speed                                                                         | 15 – 85        | rpm   |                |
| Feed Zone Temperature                                                               | 180 – 220      | °C    |                |
| Middle Zone Temperatures                                                            | 220 – 250      | °C    |                |
| Head Zone Temperature                                                               | 220 – 250      | °C    |                |
| Neck Temperature                                                                    | 220 – 250      | °C    |                |
| Cross-head Temperature                                                              | 220 – 250      | °C    |                |
| Die Temperature                                                                     | 220 – 250      | °C    |                |
| Melt Temperature                                                                    | 220 – 250      | °C    |                |
| Conductor Pre-heat Temperature                                                      | 25 – 120       | °C    |                |
| Screen Pack                                                                         | 150 – 100      | -     |                |
| Cooling Water Air Gap                                                               | 100 – 200      | mm    |                |
| Water Bath Temperature                                                              | 15 – 60        | °C    |                |

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